

Galois Covers of M_{23} over $\mathbb{Q}$: A Positive-Dimensional Approach

Braid-Orbit Reconstruction, Exact Clutching, Algebraic Collision Tails, and Six-Branch Refinements

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For $M_{23} \leq S_{23}$, the ordered passport $(2A, 2A, 3A, 5A)$ has exactly 980 generating inner Nielsen classes in one pure-braid orbit and gives a genus-119 cross-ratio cover. The all-orderings orbit has 11,760 states; a free Klein-four quotient gives a degree-2940, genus-43 reduced j -line cover with 406 cusps. The real-boundary computation certifies eight individual $\mathbb{Q}$ -rational boundary points, and $B_{27} = \mathbb{Q}[Z]/(Z^{27} - Z)$ supplies the target normal form for the interior fiber-algebra criterion.

For the five-branch passport $(2A, 2A, 2A, 2A, 3A)$, an independent computation gives a connected 21,456-state colored component. All six equal- $2A$ collision divisors have 980 width-five cycles, and admissible-cover clutching identifies every normalized width-five component with the complete genus-119 four-branch curve. The released $(2A, 23A, 23B)$ tuple admits a generating five-branch refinement. A common six-branch class $(2A^3, 3A^3)$ joins the two five-branch components through an exact 14-move colored braid word and rational collision tails. A generating $2A^{10}$ refinement has source genus 18; an exact 29,005-move B_{10} word connects its $2 + 4 + 4$ contraction to the released three-point tuple, while a separate h -equivariant ten-involution refinement fuses by five pairs into the verified five-branch component.

The mixed collision boundary is a degree-18, genus-2 cover. On the canonical ten-state cell, the central deck involution gives the phase pairing commuting with q_1 , while the native return conjugates q_1 to q_1^{-1} and yields a dihedral order-10 action. The associated set-stabilizer base change produces a degree-10 Hurwitz subcover, a degree-5 quotient of genus 161, and the comparison morphism $\mathfrak{B}_H$. The carrier $\beta_{10}(t) = 1 - (1 - t^2)^5$ quotients to $\beta_5(t) = 1 - (1 - t)^5$. The width-11 locus consists of four transitive degree-66, genus-zero components paired by an explicit inverse-return involution. Pair-clutching and balanced twisted-cover deformation complete the corresponding $\mathbb{Q}$ -defined admissible-cover geometry.

Keywords. Mathieu group M_{23} ; Hurwitz spaces; Nielsen classes; braid orbits; admissible covers; inverse Galois theory; computational group theory.